

# PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

## Module 5 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. WDM Fundamentals

### Wavelength Division Multiplexing (WDM)

**WDM** is a technology that multiplexes multiple optical carrier signals on a single fiber using different wavelengths (colors) of laser light. Each wavelength carries a separate data channel, dramatically multiplying the fiber capacity without laying additional fiber.

#### 1.1 WDM Principle and ITU-T Channel Grid

Multiple lasers operating at distinct wavelengths are combined (multiplexed) by a MUX onto a single fiber. At the far end, a DEMUX separates them back to individual receivers. Each wavelength channel is independent and carries its own data stream.

The **ITU-T G.694.1** standard defines the DWDM channel grid centered on 193.1 THz (1552.52 nm) with channel spacings of 12.5, 25, 50, or 100 GHz.

#### 1.2 CWDM vs DWDM

| Parameter         | CWDM                   | DWDM                         |
|-------------------|------------------------|------------------------------|
| Full name         | Coarse WDM             | Dense WDM                    |
| Channel spacing   | 20 nm (ITU-T G.694.2)  | 0.4–0.8 nm (50–100 GHz)      |
| Channel count     | 4–18                   | 40–160+                      |
| Wavelength range  | 1271–1611 nm           | 1530–1565 nm (C-band)        |
| Laser requirement | Uncooled (no TEC)      | Temperature-controlled (TEC) |
| Cost              | Low                    | High                         |
| Amplification     | No (too wide for EDFA) | EDFA compatible              |
| Application       | Metro, short-haul      | Long-haul, backbone          |

## 2. WDM Components

### WDM MUX/DEMUX Devices

WDM MUX combines multiple wavelengths onto one fiber; DEMUX separates them. Key technologies: Arrayed Waveguide Grating (AWG), Thin-Film Filter (TFF), and Fiber Bragg Grating (FBG).

#### 2.1 Arrayed Waveguide Grating (AWG)

An AWG is a planar integrated device with a star coupler at input, an array of waveguides of incrementally increasing lengths, and a second star coupler (phased array region). The

path-length difference between adjacent waveguides is exactly one wavelength, causing different wavelengths to focus to different output ports due to constructive interference. AWGs are used for both MUX and DEMUX.

**Advantages:** Mass-producible on a single chip; very flat passband; bidirectional; scalable to 40+ channels. **Disadvantage:** Temperature-sensitive (requires TEC for DWDM).

## 2.2 Thin-Film Filter (TFF)

Uses multiple thin-film interference layers (Fabry-Perot cavity) to reflect one wavelength while transmitting all others. Stacked in series, each filter drops one channel. **Advantage:** Low insertion loss for the reflected channel, temperature stable. **Disadvantage:** Insertion loss accumulates for large channel counts.

## 2.3 Fiber Bragg Grating (FBG)

A periodic refractive index modulation written into the fiber core. Reflects only the Bragg wavelength  $\lambda_B = 2n_{eff}\Lambda$  ( $\Lambda$  = grating period) and transmits all others. Used with circulators for wavelength drop/add.

### FBG Bragg Condition

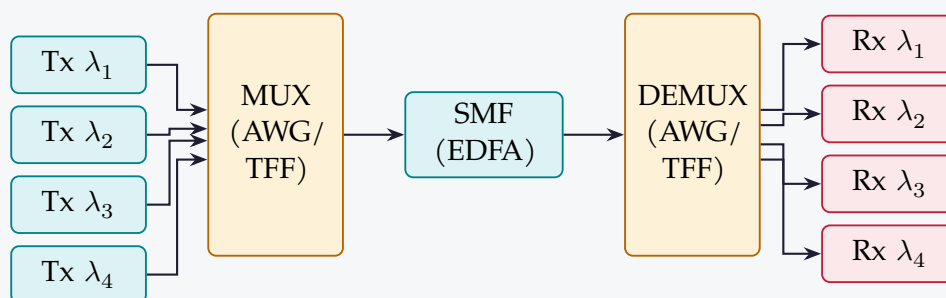
$$\lambda_B = 2n_{eff}\Lambda$$

where  $n_{eff}$  is the effective refractive index of the guided mode and  $\Lambda$  is the grating period. Tunable by temperature or strain.

## 2.4 Optical Add-Drop Multiplexer (OADM)

An OADM allows one or more wavelength channels to be dropped from and added to a WDM signal at an intermediate node without converting to electrical signals. A **ROADM** (Reconfigurable OADM) allows software-controlled selection of any channel.

### WDM System Block Diagram — 4-Channel DWDM



## 3. WDM Network Principles — OXC and Protection

### Optical Cross-Connect (OXC)

An **OXC** is a network node that switches wavelength channels from any input fiber to any output fiber, entirely in the optical domain. It is the building block of all-optical mesh networks and enables wavelength routing.

#### Mesh topology and protection:

- **Mesh topology:** Each node connects to multiple other nodes via multiple fibers. Provides multiple routing paths for each wavelength channel.
- **Protection switching:** Dedicated 1+1 protection pre-assigns a backup path. On failure, traffic is immediately switched to backup (<50 ms).
- **Restoration:** Dynamic rerouting of traffic around a failure using spare capacity. Takes longer (>100 ms) but uses network resources more efficiently.

## 4. Nonlinear Effects in Optical Fiber

### Nonlinear Effects in Fiber

At high optical power, the refractive index of silica becomes intensity-dependent:  $n = n_0 + n_2 I$ , where  $n_2 \approx 2.6 \times 10^{-20} \text{ m}^2/\text{W}$  is the nonlinear refractive index coefficient. This gives rise to three key nonlinear phenomena: SPM, XPM, and FWM.

#### 4.1 Self-Phase Modulation (SPM)

A pulse modulates its own phase due to the intensity-dependent refractive index. The nonlinear phase shift accumulated is:

$$\phi_{NL} = \gamma P_0 L_{eff}, \quad \gamma = \frac{n_2 \omega}{c A_{eff}} = \frac{2\pi n_2}{\lambda A_{eff}}$$

where  $\gamma$  is the fiber nonlinearity coefficient,  $P_0$  is peak power,  $L_{eff}$  is the effective length.

**B-integral** (total accumulated nonlinear phase):

$$B = \gamma P_0 L_{eff}$$

SPM causes **spectral broadening** of the pulse. The leading edge of the pulse is red-shifted; the trailing edge is blue-shifted. In the anomalous dispersion regime, SPM and dispersion can balance to form solitons.

#### 4.2 Cross-Phase Modulation (XPM)

In a WDM system, the intensity of one channel modulates the phase of an adjacent channel. The nonlinear phase induced in channel 2 by channel 1 is  $2\gamma P_1 L_{eff}$  (factor of 2 due to polarization mixing). XPM causes amplitude-to-phase conversion, broadens spectra, and creates timing jitter in WDM channels.

### 4.3 Four-Wave Mixing (FWM)

When three channels at frequencies  $\omega_1, \omega_2, \omega_3$  co-propagate, a new frequency  $\omega_4 = \omega_1 + \omega_2 - \omega_3$  is generated (and other combinations). This creates crosstalk between WDM channels.

#### FWM Phase Matching Condition

FWM is most efficient when the **phase mismatch**  $\Delta\beta$  is small:

$$\Delta\beta = \beta(\omega_1) + \beta(\omega_2) - \beta(\omega_3) - \beta(\omega_4) \approx 0$$

In terms of the dispersion parameter  $D$ :

$$\Delta\beta \approx \frac{2\pi c D}{\lambda^2} \Delta\lambda^2 \cdot \lambda^2$$

FWM is strongest near the zero-dispersion wavelength (small  $D$ ) and for equal channel spacing. Mitigation: use non-zero dispersion-shifted fiber (NZ-DSF) with  $D \neq 0$  in the signal band.

## 5. Group Velocity Dispersion (GVD) and Dispersion Parameter

#### GVD and Dispersion Parameter $D$

**Group velocity dispersion (GVD)** refers to the variation of group velocity with frequency, parameterized by  $\beta_2 = d^2\beta/d\omega^2$  (ps<sup>2</sup>/km). The **dispersion parameter**  $D$  relates directly to  $\beta_2$ :  $D = -\frac{2\pi c}{\lambda^2} \beta_2$  (ps/nm·km).

- **Normal dispersion:**  $\beta_2 > 0$  ( $D < 0$ ); shorter wavelengths travel faster; pulse broadens normally.
- **Anomalous dispersion:**  $\beta_2 < 0$  ( $D > 0$ ); longer wavelengths travel faster; enables soliton formation.
- **Zero-dispersion wavelength  $\lambda_0$ :** For standard SMF,  $\lambda_0 \approx 1310$  nm. In DSF, shifted to 1550 nm.

#### Dispersion Length

$$L_D = \frac{T_0^2}{|\beta_2|}$$

where  $T_0$  is the initial pulse half-width (1/e point).  $L_D$  is the length at which the pulse broadens by  $\sqrt{2}$ . For  $L \ll L_D$ , dispersion is negligible.

## 6. Soliton-Based Communication

### Optical Soliton

An **optical soliton** is a pulse that propagates through a dispersive fiber without changing its shape, because the effects of anomalous GVD (pulse spreading) are exactly balanced by SPM (pulse compression). Solitons are solutions to the Nonlinear Schrödinger Equation (NLSE).

#### 6.1 Nonlinear Schrödinger Equation (NLSE)

The propagation of an optical pulse in a fiber is governed by:

$$j\frac{\partial A}{\partial z} - \frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + \gamma|A|^2A = 0$$

where  $A(z, T)$  is the slowly-varying pulse envelope,  $\beta_2 < 0$  for anomalous dispersion, and  $\gamma$  is the nonlinear coefficient.

#### 6.2 Fundamental Soliton ( $N = 1$ )

The fundamental soliton has the **sech** profile:

$$A(0, T) = \sqrt{P_0} \text{sech}(T/T_0)$$

The soliton condition relates dispersion and nonlinear lengths:

#### Soliton Condition and Width-Power Relation

$$L_D = L_{NL}, \quad \text{i.e.,} \quad \frac{T_0^2}{|\beta_2|} = \frac{1}{\gamma P_0}$$

$$\Rightarrow P_0 = \frac{|\beta_2|}{\gamma T_0^2}$$

The peak power required for a soliton increases as the pulse width  $T_0$  decreases. For  $T_0 = 10 \text{ ps}$ ,  $|\beta_2| = 20 \text{ ps}^2/\text{km}$ ,  $\gamma = 2 \text{ W}^{-1}\text{km}^{-1}$ :  $P_0 = 20/(2 \times 100) = 0.1 \text{ W}$ .

Higher-order solitons ( $N = 2, 3, \dots$ ) periodically breathe (compress and expand) with soliton period  $z_s = \pi L_D/2$ .

#### 6.3 Advantages and Limitations of Soliton Communication

##### Advantages:

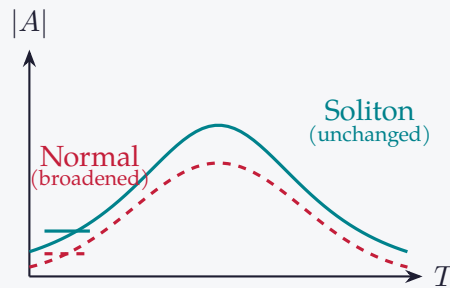
- Dispersion-free propagation: pulse shape preserved over trans-oceanic distances.
- Allows very high bit rates (sub-picosecond pulses).
- With periodic amplification, solitons can be maintained over global distances.

##### Limitations:

- Requires anomalous dispersion ( $D > 0$ ) throughout the link; problematic with DCF.
- Gordon-Haus jitter: ASE noise from amplifiers causes random timing jitter, limiting transmission distance.

- Soliton–soliton interaction: adjacent pulses at the same wavelength must be spaced  $> 5T_0$  apart.
- Requires precise control of launch power and pulse shape.
- Dispersion management (DM-solitons) is needed for WDM compatibility.

#### Soliton Propagation vs Normal Pulse Propagation



## Quick Revision — Module 5

WDM Systems and Nonlinear Effects | PE-EC801B

### Key Formulas at a Glance

|                                  |                                                                           |
|----------------------------------|---------------------------------------------------------------------------|
| <b>FBG Bragg condition</b>       | $\lambda_B = 2n_{eff}\Lambda$                                             |
| <b>SPM nonlinear phase</b>       | $\phi_{NL} = \gamma P_0 L_{eff}$ (B-integral)                             |
| <b>Fiber nonlinearity coeff.</b> | $\gamma = 2\pi n_2 / (\lambda A_{eff})$ ( $\text{W}^{-1}\text{km}^{-1}$ ) |
| <b>Dispersion parameter</b>      | $D = -(2\pi c / \lambda^2) \beta_2$ ( $\text{ps/nm}\cdot\text{km}$ )      |
| <b>Dispersion length</b>         | $L_D = T_0^2 /  \beta_2 $                                                 |
| <b>Nonlinear length</b>          | $L_{NL} = 1 / (\gamma P_0)$                                               |
| <b>Soliton condition</b>         | $L_D = L_{NL}$                                                            |
| <b>Soliton peak power</b>        | $P_0 =  \beta_2  / (\gamma T_0^2)$                                        |
| <b>Soliton shape</b>             | $A(0, T) = \sqrt{P_0} \text{sech}(T/T_0)$                                 |
| <b>DWDM channel spacing</b>      | 50–100 GHz (ITU-T G.694.1)                                                |

### One-Line Definitions

- **WDM:** Multiple wavelength channels on a single fiber; multiplies capacity without new fiber.
- **DWDM:** Dense WDM with 50/100 GHz channel spacing; supports 40–160+ channels in C-band.
- **AWG:** Arrayed Waveguide Grating; planar integrated MUX/DEMUX using path-length differences.
- **FBG:** Fiber Bragg Grating; periodic index modulation reflecting only the Bragg wavelength.
- **OADM:** Optical Add-Drop Multiplexer; drops and adds wavelength channels at intermediate nodes.
- **OXC:** Optical Cross-Connect; switches wavelengths between fibers in an all-optical network node.
- **SPM:** Self-Phase Modulation; pulse modulates its own phase, causing spectral broadening.
- **XPM:** Cross-Phase Modulation; one WDM channel modulates the phase of another.
- **FWM:** Four-Wave Mixing; three channels generate a fourth frequency; crosstalk near  $\lambda_0$ .
- **Soliton:** Pulse that propagates undistorted due to exact SPM–GVD balance (anomalous dispersion).

- $\beta_2$ : GVD parameter;  $\beta_2 < 0$  = anomalous;  $\beta_2 > 0$  = normal dispersion.

### Frequently Asked / Viva Questions

#### Q1. What is WDM and what are its advantages?

WDM multiplexes multiple wavelength channels on one fiber. Advantages: multiplies capacity  $N$ -fold without laying new fiber; each channel is protocol- and bit-rate-independent; can combine CWDM and DWDM in hybrid networks.

#### Q2. Differentiate between CWDM and DWDM.

CWDM: 20 nm channel spacing, 4–18 channels, 1271–1611 nm, uncooled lasers, low cost, no EDFA. DWDM: 50–100 GHz spacing, 40–160+ channels, 1530–1565 nm, TEC-cooled lasers, EDFA-compatible, high capacity, high cost.

#### Q3. What is an AWG and how does it work?

An Arrayed Waveguide Grating is a planar integrated optics device. Light enters a free propagation region (star coupler), splits into an array of waveguides with linearly increasing lengths, recombines in a second star coupler. The path-length differences cause different wavelengths to constructively interfere at different output ports.

#### Q4. What is an FBG? State the Bragg condition.

A Fiber Bragg Grating is a periodic modulation of the refractive index written into the fiber core using UV light. It reflects only  $\lambda_B = 2n_{eff}\Lambda$  and transmits all others. Used for channel dropping in OADMs, dispersion compensation, and sensing.

#### Q5. What is Self-Phase Modulation? What does it cause?

SPM arises because  $n = n_0 + n_2 I$ . A pulse modulates its own phase:  $\phi_{NL}(t) = \gamma P(t) L_{eff}$ . The leading edge is red-shifted and trailing edge is blue-shifted, causing spectral broadening. In anomalous dispersion, SPM enables soliton formation.

#### Q6. What is Four-Wave Mixing and when is it worst?

FWM generates new frequencies  $\omega_4 = \omega_1 + \omega_2 - \omega_3$  from three co-propagating channels. It is worst when the phase mismatch  $\Delta\beta \approx 0$ , i.e., near the zero-dispersion wavelength  $\lambda_0$ . Equal channel spacing also maximizes FWM efficiency. NZ-DSF ( $D \neq 0$ ) mitigates it.

#### Q7. Define the dispersion parameter $D$ and state its units.

$D = -(2\pi c/\lambda^2)\beta_2$  in units of ps/nm·km. It gives the pulse broadening per nm of source linewidth per km of fiber. Standard SMF:  $D \approx +17$  ps/nm·km at 1550 nm.

#### Q8. What is anomalous dispersion and why is it needed for solitons?

Anomalous dispersion ( $\beta_2 < 0$ ,  $D > 0$ ): longer wavelengths travel faster. The chirp induced by anomalous GVD (red leading) is opposite to SPM chirp (red leading). At the right power, these two effects exactly cancel, producing a shape-invariant soliton.

## Frequently Asked / Viva Questions (contd.)

**Q9. State the soliton condition and derive the soliton power.**

Soliton condition:  $L_D = L_{NL}$ , i.e.,  $T_0^2/|\beta_2| = 1/(\gamma P_0)$ . Solving:  $P_0 = |\beta_2|/(\gamma T_0^2)$ .  
For  $T_0 = 10$  ps,  $|\beta_2| = 20$  ps<sup>2</sup>/km,  $\gamma = 2$  W<sup>-1</sup>km<sup>-1</sup>:  $P_0 = 0.1$  W.

**Q10. What are the limitations of soliton communication?**

Gordon-Haus timing jitter (from ASE noise), soliton-soliton interaction requiring large pulse spacing ( $> 5T_0$ ), requirement for precisely anomalous dispersion throughout, and incompatibility with standard DWDM channels without dispersion-managed fiber.

**Q11. What is XPM and how does it differ from SPM?**

XPM: the intensity of channel  $i$  modifies the phase of channel  $j$  with a coefficient of  $2\gamma$  (vs  $\gamma$  for SPM). XPM causes amplitude-dependent phase noise and timing jitter in WDM systems, particularly at high channel counts.

**Q12. What is an OADM and what is its purpose in WDM networks?**

An Optical Add-Drop Multiplexer allows one or more wavelength channels to be dropped to and added from a local node while the remaining channels pass through transparently. A ROADM adds software reconfigurability, supporting any-to-any wavelength routing.

## Nonlinear Effects — Summary Comparison

| Effect | Mechanism                       | Impact                            | Mitigation                 |
|--------|---------------------------------|-----------------------------------|----------------------------|
| SPM    | $n_2 I$ on own pulse            | Spectral broadening               | Reduce power; use DSF      |
| XPM    | $n_2 I$ from other channel      | Phase noise, jitter               | Channel spacing, filtering |
| FWM    | 3-wave mixing $\rightarrow$ 4th | Crosstalk; power drain            | NZ-DSF; unequal spacing    |
| SRS    | Raman scattering                | Power transfer to lower $\lambda$ | Limit span power           |